Kautik Rehani

BS–MS student, Physics major and Mathematics minor, IISER Pune
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Education

Indian Institute of Science Education and Research, Pune Expected July 2028
BS–MS Dual Degree; Physics major, Mathematics minor

Research Experience

Long-term Project Student, IISER Pune Aug 2025–present
Supervisor: Prof. Sudarshan Ananth

- Worked through three-point interaction terms for general spin and the quartic gravity interaction in terms of the two graviton helicity states in light-cone gauge.
- Worked out the full Yang-Mills Lagrangian in light-cone gauge; investigated higher-order terms in closed-form gravity expressions with Dr. Nipun Bhave.
- Current reading: Color-kinematics duality and $\mathcal{N} = 8$ supergravity in physical modes.

Summer Intern, NCRA-TIFR, Pune May - Jun 2025
Supervisor: Prof. Ishwara Chandra

- Worked on code to identify changing-look AGN candidates in large datasets, using SDSS Data Releases 16 and 17.
- Studied changing-look AGNs and their connection to black-hole accretion.

Summer Intern, TIFR, Mumbai May - Jul 2024
Supervisor: Prof. Shadab Alam

- Worked on QSO spectral stacking within the broader project *QSOs as tracers of our Universe.*
- Studied spectral analysis and explored black-hole kick-velocity distributions.

Relevant Coursework

QFT-1 — Prof. Diptimoy Ghosh
Classical field theory, symmetries, free-field quantisation, interacting theories and scattering.

Differential Geometry and Lie Groups — Prof. Tejas Kalelkar
Smooth manifolds, differential forms, integration, Lie groups, Lie algebras and group actions.

Gravitation and Quantum Entanglement — Prof. Sunil Mukhi
Entanglement in QFT, modular Hamiltonians, black-hole thermodynamics and holographic entropy.

Technical Skills & Languages

Programming and computing: Python, TensorFlow; basic Mathematica, SageMath and Gnuplot.
Tools: LaTeX, GitHub/GitLab, Linux/Ubuntu, Jupyter Notebook, Anaconda.
Languages: English and Hindi (fluent); French (basic reading and writing).

Achievements

NEST 2023: All India Rank 30.
KVPY-SA 2021: All India Rank 1557.

Activities & Involvement

Akashganga: Astronomy Club member; telescope-building and cosmology workshops.
Disha: Volunteer in science outreach for local school students.
Smashers: Founded an institute sports community with approximately 200 members.